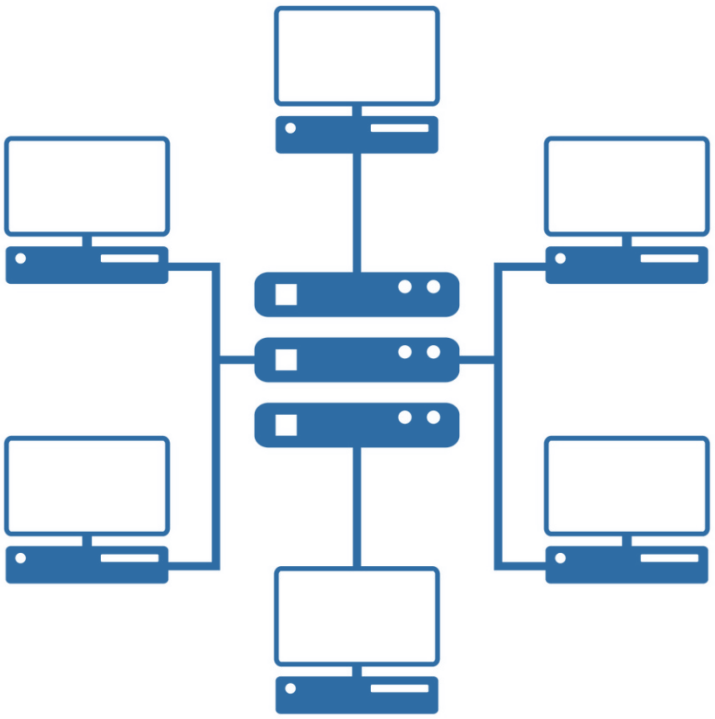
2022-2023

Computer Networks

By GTU Medium



COMPUTER ENGINEERING DEPARTMENT

SUBJECT: - COMPUTER NETWORK (3150701)

SEMESTER:-5th

QUESTION BANK

1. Explain functionality of Repeater, Hub, Bridge, Switch, Router and Gateway.

Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network.

Hub – A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

Routers- Routers are networking devices operating at layer 3 or a network layer of the OSI model. They are responsible for receiving, analyzing, and forwarding data packets among the connected computer networks. When a data packet arrives, the router inspects the destination address, consults its routing tables to decide the optimal route and then transfers the packet along this route.

Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take

data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

2. What is Computer Network? List out the classification of Network.

A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.

A computer network can be categorized by their size. A computer network is mainly of four types:

1. LAN(Local Area Network)
2. PAN(Personal Area Network)
3. MAN(Metropolitan Area Network)
4. WAN(Wide Area Network)

LAN(Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.



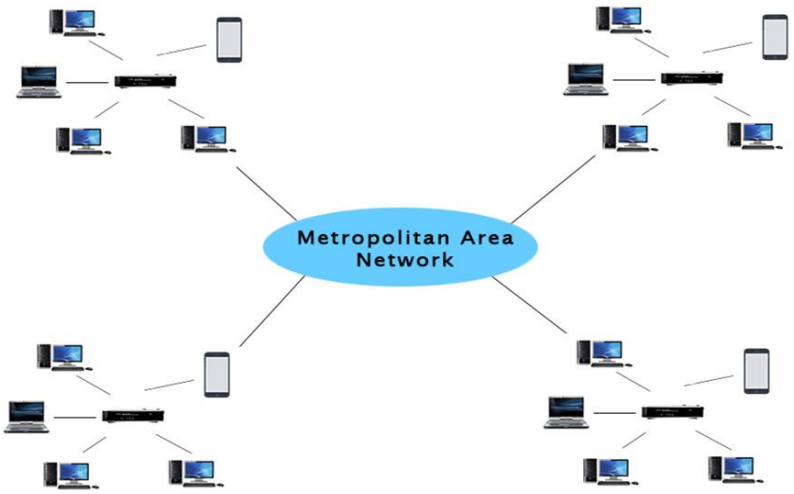
PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.



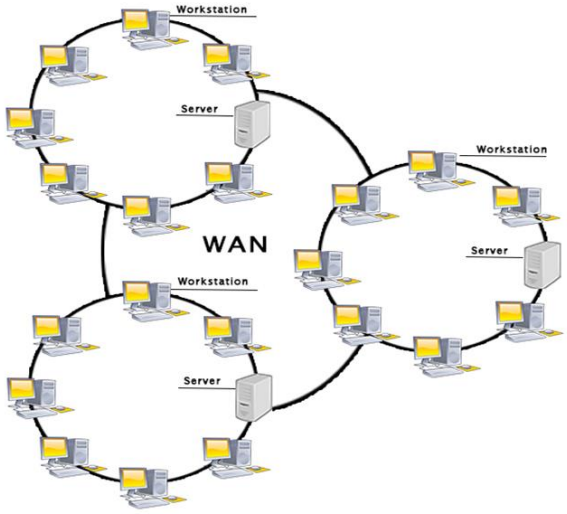
MAN(Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network(LAN).



WAN(Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.



3. What is Protocol?

A network protocol is an established set of rules that determine how data is transmitted between different devices in the same network. Essentially, it allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure or design. Network protocols are the reason you can easily communicate with people all over the world, and thus play a critical role in modern digital communications.

4. What is the Network Edge?

The network edge is generally defined as the place where a device or local area network (LAN) connects to the Internet. It is called the “edge,” because this is the entry point to the network where devices themselves are communicating with the Internet.

5. Difference between circuit switching and packet switching.

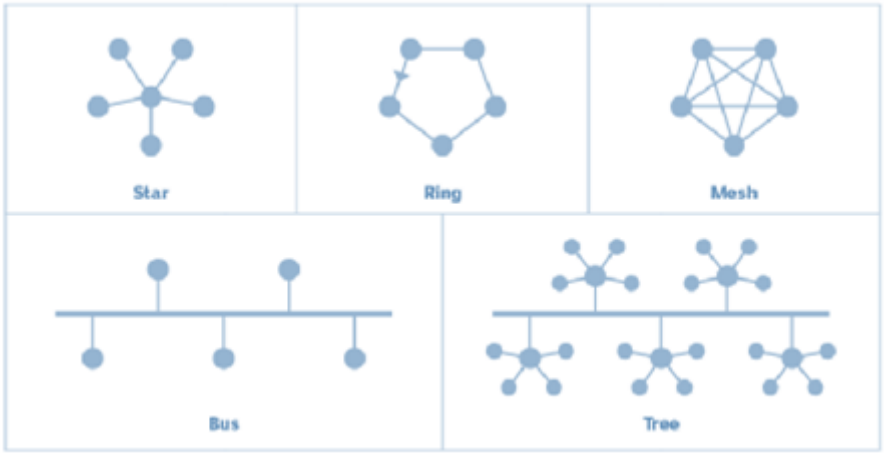
Circuit Switching	Packet Switching
Circuit switching requires a dedicated path before sending data from source to destination.	Packet switching does not require any dedicated path to send data from source to destination.
It reserves the entire bandwidth in advance.	It does not reserve bandwidth in advance
No store and forward transmission	It supports store and forward transmission
Each packet follows the same route	A packet can follow any route
Call setup is required	No call setup is required
Bandwidth wastage	No bandwidth wastage

*In telecommunication, call setup is the process of establishing a virtual circuit across a telecommunications network.

*Bandwidth- The maximum amount of data transmitted over an internet connection in a given amount of time.

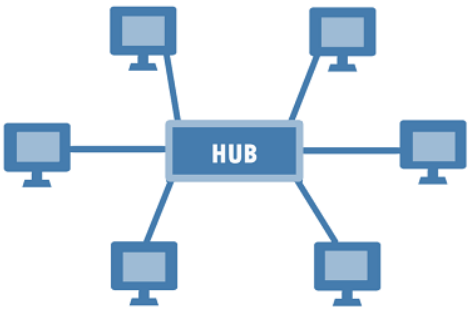
6. What is topology? Explain star topology in brief.

A network topology is the physical and logical arrangement of nodes and connections in a network. Nodes usually include devices such as switches, routers and software with switch and router features. Network topologies are often represented as a graph.



Star Topology:

Star topology is a network topology in which each network component is physically connected to a central node such as a router, hub or switch.

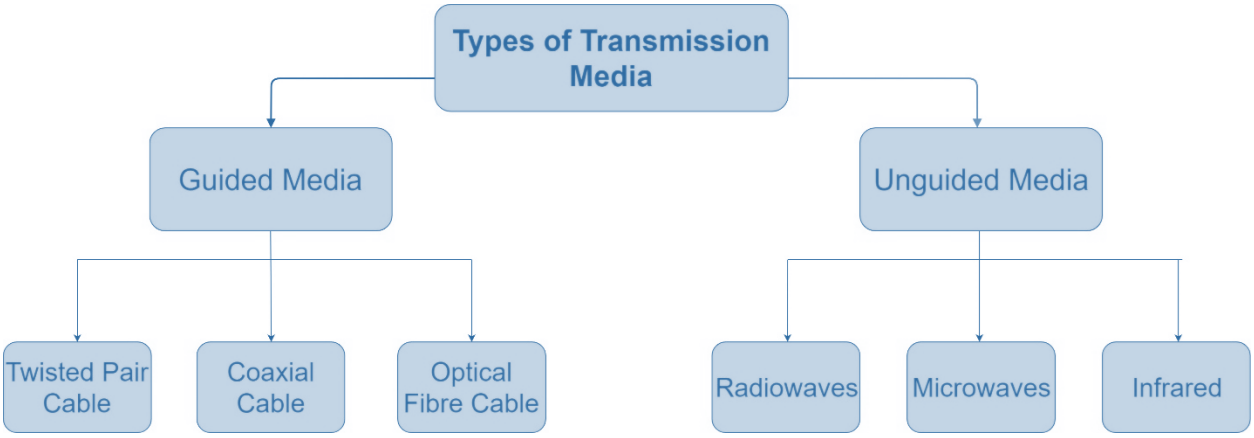


A device cannot provide communication directly between the devices in a star topology; it has to communicate with the help of a hub. The peripheral devices function as clients, while the central network device serves as a server.

In star topology, all connected devices are completely dependent on the central device; the communication through the whole Computer Network fails if the central device gets any problem.

7. List out Transmission Media. Explain anyone.

In data communication terminology, a transmission medium is a physical path between the transmitter and the receiver i.e. it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into the following types:



1. Guided Media:
It is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Features:

- High Speed
- Secure
- Used for comparatively shorter distances

There are 3 major types of Guided Media:

(i) Twisted Pair Cable –
It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. Twisted Pair is of two types:

1. **Unshielded Twisted Pair (UTP):**
   This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages:

- Least expensive
- Easy to install
- High speed capacity

Disadvantages:

- Susceptible to external interference
- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

2. **Shielded Twisted Pair (STP):**
   This type of cable consists of a special jacket to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages:

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparitively faster

Disadvantages:

- Comparitively difficult to install and manufacture
- More expensive
- Bulky

(ii) Coaxial Cable –
It has an outer plastic covering containing 2 parallel conductors each having a separate insulated protection cover. Coaxial cable transmits information in two modes: Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.

Advantages:

- High Bandwidth
- Better noise Immunity
- Easy to install and expand

- Inexpensive

Disadvantages:

- Single cable failure can disrupt the entire network

(iii) Optical Fibre Cable –
It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data.

Advantages:

- Increased capacity and bandwidth
- Light weight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages:

- Difficult to install and maintain
- High cost
- Fragile
- unidirectional, ie, will need another fibre, if we need bidirectional communication

2. Unguided Media:
It is also referred to as Wireless or Unbounded transmission media.No physical medium is required for the transmission of electromagnetic signals.

Features:

- Signal is broadcasted through air
- Less Secure
- Used for larger distances

There are 3 major types of Unguided Media:

(i) Radiowaves –
These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range: 3KHz – 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.

Further Categorized as (i) Terrestrial and (ii) Satellite.

(ii) Microwaves –
It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz. These are majorly used for mobile phone communication and television distribution.

(iii) Infrared –
Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range: 300GHz – 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.

8. Discuss throughput in the network.

In data transmission, network throughput is the amount of data moved successfully from one place to another in a given time period, and typically measured in bits per second (bps), as in megabits per second (Mbps) or gigabits per second (Gbps).

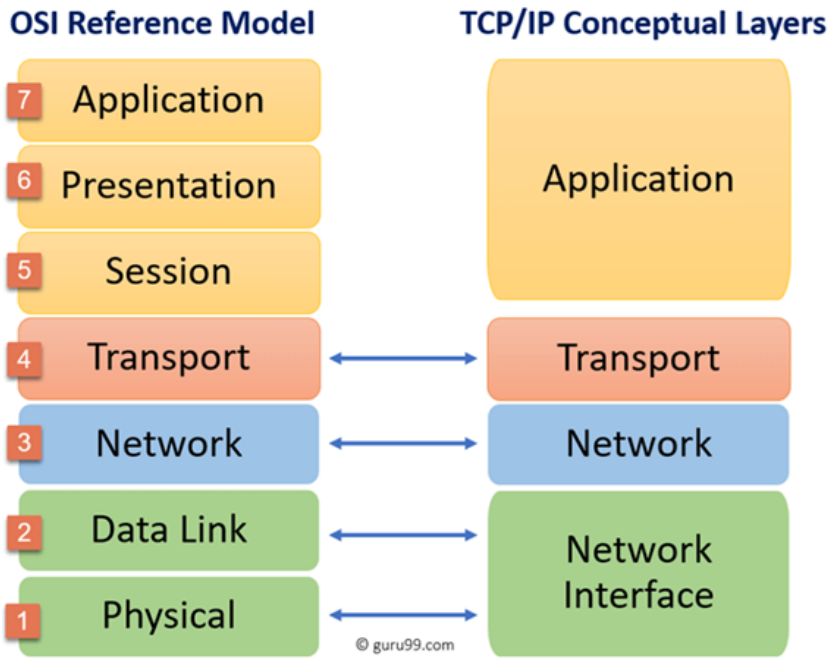


9. Differentiate TCP/IP protocol stack and OSI Reference model of the computer network.

OSI	TCP/IP
OSI represents Open System Interconnection.	TCP/IP model represents the Transmission Control Protocol / Internet Protocol.

OSI	TCP/IP
OSI is a generic, protocol independent standard. It is acting as an interaction gateway between the network and the final-user.	TCP/IP model depends on standard protocols about which the computer network has created. It is a connection protocol that assigns the network of hosts over the internet.
The OSI model was developed first, and then protocols were created to fit the network architecture's needs.	The protocols were created first and then built the TCP/IP model.
It provides quality services.	It does not provide quality services.
In OSI model the transport layer guarantees the delivery of packets.	In TCP/IP model the transport layer does not guarantees delivery of packets. Still the TCP/IP model is more reliable.
Protocols are hidden in OSI model and are easily replaced as the technology changes.	The TCP/IP model protocols are not hidden, and we cannot fit a new protocol stack in it.
It is difficult as distinguished to TCP/IP.	It is simpler than OSI.
It provides both connection and connectionless oriented transmission in the network layer; however, only connection-oriented transmission in the transport layer.	It provides connectionless transmission in the network layer and supports connecting and connectionless-oriented transmission in the transport layer.
It uses a horizontal approach.	It uses a vertical approach.
The smallest size of the OSI header is 5 bytes.	The smallest size of the TCP/IP header is 20 bytes.

OSI	TCP/IP
Protocols are unknown in the OSI model and are returned while the technology modifies.	In TCP/IP, returning protocol is not difficult.



10. Explain various delay which are occur in data packet transmission.

1. Transmission Delay:

The time taken to transmit a packet from the host to the transmission medium is called Transmission delay.



2. Propagation delay:
After the packet is transmitted to the transmission medium, it has to go through the medium to reach the destination. Hence the time taken by the last bit of the packet to reach the destination is called propagation delay.



3. Queueing delay:
Let the packet is received by the destination, the packet will not be processed by the destination immediately. It has to wait in a queue in something called a buffer. So the amount of time it waits in queue before being processed is called queueing delay.

4. Processing delay:
Now the packet will be taken for the processing which is called processing delay.

Time is taken to process the data packet by the processor that is the time required by intermediate routers to decide where to forward the packet, update TTL, perform header checksum calculations.

11. Describe World Wide Web.

World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media to your device.

12. What is Domain Name System? Explain it with resource records.

DNS is a core internet technology that translates human-friendly domain names into machine-usable IP addresses, such as www.example.com into 192.0. 2.1.

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.

- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.
- DNS is a client/server network communication protocol. DNS clients send requests to the. server while DNS servers send responses to the client. Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.

Resource records are used to store data about domain names and IP addresses.

A DNS zone database is made up of a collection of resource records. Each resource record specifies information about a particular object.

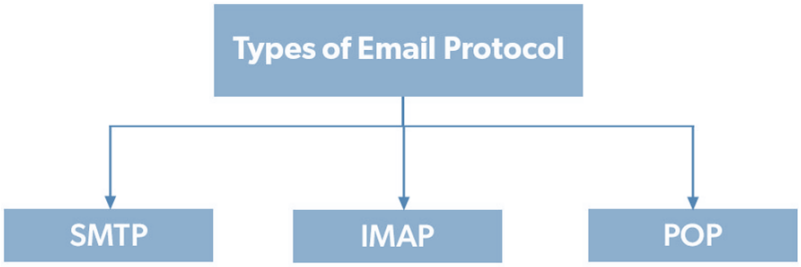
For example, address mapping (A) records map a host name to an IP address, and reverse-lookup pointer (PTR) records map an IP address to a host name. The server uses these records to answer queries for hosts in its zone.

13. Explain Email and email protocols.

Electronic Mail (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a message in formatted manner (mail) to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data.

Email protocol is a set of rules defined to ensure that emails can be exchanged between various servers and email clients in a standard manner. This ensures that the email is universal and works for all users.

Three basic types of email protocols involved for sending and receiving mails are:



SMTP (Simple Mail Transfer Protocol):

Simple Mail Transfer Protocol is used to send mails over the internet. SMTP is an application layer and connection-oriented protocol. SMTP is efficient and reliable for sending emails. SMTP uses TCP as the transport layer protocol. It handles the sending and receiving of messages between email servers over a TCP/IP network. This protocol along with sending emails also provides the feature of notification for incoming mails. When a sender sends an email then the sender's mail client sends it to the sender's mail server and then it is sent to the receiver mail server through SMTP. SMTP commands are used to identify the sender and receiver email addresses along with the message to be sent.

Some of the SMTP commands are HELLO, MAIL FROM, RCPT TO, DATA, QUIT, VERIFY, SIZE, etc. SMTP sends an error message if the mail is not delivered to the receiver hence, reliable protocol.

POP(Post Office Protocol):

Post Office Protocol is used to retrieve email for a single client. POP3 version is the current version of POP used. It is an application layer protocol. It allows to access mail offline and thus, needs less internet time. To access the message it has to be downloaded. POP allows only a single mailbox to be created on the mail server. POP does not allow search facilities

Some of the POP commands are LOG IN, STAT, LIST, RETR, DELE, RSET, and QUIT.

IMAP(Internet Message Access Protocol):

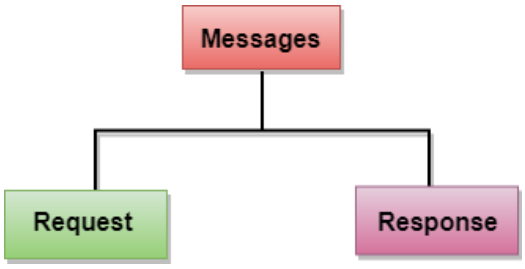
Internet Message Access Protocol is used to retrieve mails for multiple clients. There are several IMAP versions: IMAP, IMAP2, IMAP3, IMAP4, etc. IMAP is an application layer protocol. IMAP allows to access email without downloading them and also supports email download. The emails are maintained by the remote server. It enables all email operations such as creating, manipulating, delete the email without reading it. IMAP

allows you to search emails. It allows multiple mailboxes to be created on multiple mail servers and allows concurrent access. Some of the IMAP commands are: IMAP_LOGIN, CREATE, DELETE, RENAME, SELECT, EXAMINE, and LOGOUT.

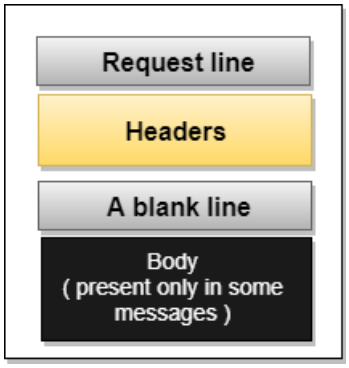
14. Describe HYPER TEXT TRANSFER PROTOCAL.

- HTTP stands for **HyperText Transfer Protocol**.
- It is a protocol used to access the data on the World Wide Web (www).
- The HTTP protocol can be used to transfer the data in the form of plain text, hypertext, audio, video, and so on.
- This protocol is known as HyperText Transfer Protocol because of its efficiency that allows us to use in a hypertext environment where there are rapid jumps from one document to another document.
- HTTP is similar to the FTP as it also transfers the files from one host to another host. But, HTTP is simpler than FTP as HTTP uses only one connection, i.e., no control connection to transfer the files.
- HTTP is used to carry the data in the form of MIME-like format.
- HTTP is similar to SMTP as the data is transferred between client and server. The HTTP differs from the SMTP in the way the messages are sent from the client to the server and from server to the client. SMTP messages are stored and forwarded while HTTP messages are delivered immediately.

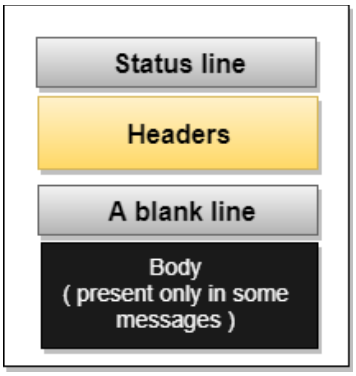
HTTP messages are of two types: request and response. Both the message types follow the same message format.



Request Message: The request message is sent by the client that consists of a request line, headers, and sometimes a body.



Response Message: The response message is sent by the server to the client that consists of a status line, headers, and sometimes a body.



15. What is transport layer? Explain in brief.

Transport layer offers peer-to-peer and end-to-end connection between two processes on remote hosts. Transport layer takes data from upper layer (i.e. Application layer) and then breaks it into smaller size segments, numbers each byte, and hands over to lower layer (Network Layer) for delivery.

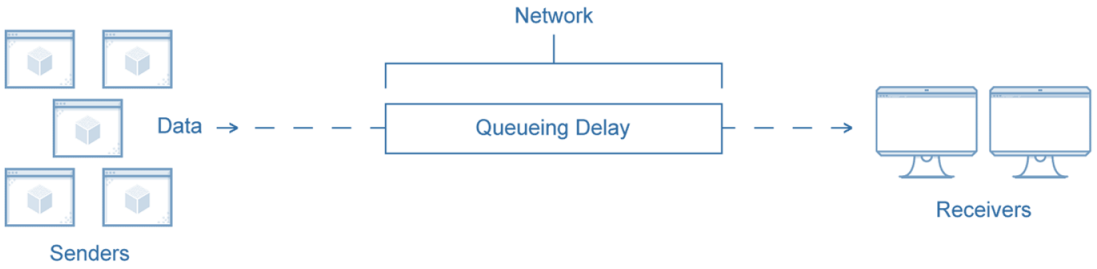
The transport layer takes services from the Network layer and provides services to the Application layer

At the sender's side: The transport layer receives data (message) from the Application layer and then performs Segmentation, divides the actual message into segments, adds source and destination's port numbers into the header of the segment, and transfers the message to the Network layer.

At the receiver's side: The transport layer receives data from the Network layer, reassembles the segmented data, reads its header, identifies the port number, and forwards the message to the appropriate port in the Application layer.

16. Explain TCP Congestion control in detail.

Network congestion refers to a reduction in quality of service (QOS) that causes packet loss, queueing delay, or the blocking of new connections. Typically, network congestion occurs in cases of traffic overloading when a link or network node is handling data in excess of its capacity.



TCP uses a congestion window and a congestion policy that avoid congestion. Previously, we assumed that only the receiver can dictate the sender's window size. We ignored another entity here, the network. If the network cannot deliver the data as fast as it is created by the sender, it must tell the sender to slow down. In other words, in addition to the receiver, the network is a second entity that determines the size of the sender's window.

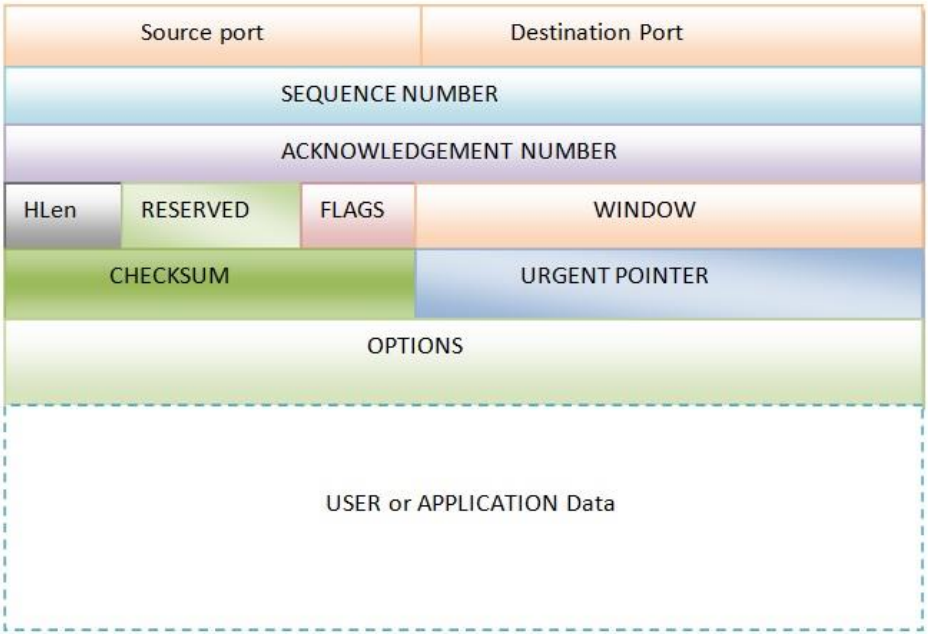
Congestion policy in TCP –

1. Slow Start Phase: starts slowly increment is exponential to threshold
2. Congestion Avoidance Phase: After reaching the threshold increment is by 1
3. Congestion Detection Phase: Sender goes back to Slow start phase or Congestion avoidance phase.

17. Explain the header format of TCP.

- **Source Port Address –**
  A 16-bit field that holds the port address of the application that is sending the data segment.
- **Destination Port Address –**
  A 16-bit field that holds the port address of the application in the host that is receiving the data segment.

- **Sequence Number –**
  A 32-bit field that holds the sequence number, i.e, the byte number of the first byte that is sent in that particular segment. It is used to reassemble the message at the receiving end of the segments that are received out of order.



- **Acknowledgement Number –**
  A 32-bit field that holds the acknowledgement number, i.e, the byte number that the receiver expects to receive next. It is an acknowledgement for the previous bytes being received successfully.
- **Header Length (HLEN) –**
  This is a 4-bit field that indicates the length of the TCP header by a number of 4-byte words in the header, i.e if the header is 20 bytes(min length of TCP header), then this field will hold 5 (because 5 x 4 = 20) and the maximum length: 60 bytes, then it'll hold the value 15(because 15 x 4 = 60). Hence, the value of this field is always between 5 and 15.
- **Control flags –**
  These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc. Their function is:
    - URG: Urgent pointer is valid

- ACK: Acknowledgement number is valid(used in case of cumulative acknowledgement)
- PSH: Request for push
- RST: Reset the connection
- SYN: Synchronize sequence numbers
- FIN: Terminate the connection

- **Window size –**
  This field tells the window size of the sending TCP in bytes.
- **Checksum –**
  This field holds the checksum for error control. It is mandatory in TCP as opposed to UDP.
- **Urgent pointer –**
  This field (valid only if the URG control flag is set) is used to point to data that is urgently required that needs to reach the receiving process at the earliest. The value of this field is added to the sequence number to get the byte number of the last urgent byte.

18. What is the different between TCP and UDP?

Key	TCP	UDP
Definition	It is a communications protocol, using which the data is transmitted between systems over the network.In this, the data is transmitted in the form of packets. It includes error-checking, guarantees the delivery and preserves the order of the data packets.	It is same as the TCP protocol except this doesn't guarantee the error-checking and data recovery. If you use this protocol, the data will be sent continuously, irrespective of the issues in the receiving end.
Design	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.

Reliability	TCP is more reliable as it provides error checking support and also guarantees delivery of data to the destination router.	UDP, on the other hand, provides only basic error checking support using checksum. So, the delivery of data to the destination cannot be guaranteed in UDP as in case of TCP.
Data transmission	In TCP, the data is transmitted in a particular sequence which means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP in order to implement ordering it has to be managed by the application layer.
Performance	TCP is slower and less efficient in performance as compared to UDP. Also TCP is heavy-weight as compared to UDP.	UDP is faster and more efficient than TCP.
Retransmission	Retransmission of data packets is possible in TCP in case packet get lost or need to resend.	Retransmission of packets is not possible in UDP.
Sequencing	The Transmission Control Protocol has a function that allows data to be sequenced (TCP). This implies that packets arrive at the recipient in the sequence they were sent.	In UDP, there is no data sequencing. The application layer must control the order if it is necessary.
Header size	TCP uses a variable-length (20-60) bytes header.	UDP has a fixed-length header of 8 bytes.
Handshake	Handshakes such as SYN, ACK, and SYNACK are used.	It's a connectionless protocol, which means it doesn't require a handshake.

Broadcasting	Broadcasting is not supported by TCP.	Broadcasting is supported by UDP.
Examples	HTTP, HTTPs, FTP, SMTP, and Telnet use TCP.	DNS, DHCP, TFTP, SNMP, RIP, and VoIP use UDP.

19. Explain Multiplexing and demultiplexing.

Multiplexing and Demultiplexing services are provided in almost every protocol architecture ever designed. UDP and TCP perform the demultiplexing and multiplexing jobs by including two special fields in the segment headers: the source port number field and the destination port number field.

Multiplexing

Multiplexing is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver.

- In Multiplexing at the Transport Layer, the data is collected from various application processes. These segments contain the source port number, destination port number, header files, and data.
- These segments are passed to the Network Layer which adds the source and destination IP address to get the datagram.

Demultiplexing

Delivering the received segments at the receiver side to the correct app layer processes is called demultiplexing.

- The destination host receives the IP datagrams; each datagram has a source IP address and a destination IP address.
- Each datagram carries 1 transport layer segment.
- Each segment has the source and destination port number.
- The destination host uses the IP addresses and port numbers to direct the segment to the appropriate socket.